BAVITHA RAAVI

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Professional Summary

Data Engineer with 5+ years of experience building batch and streaming data pipelines in technology operations, healthcare, and banking. Built Spark, BigQuery, and Airflow pipelines processing ∼50 GB of daily data at Google, consolidated data from 300+ dialysis clinics at U.S. Renal Care, and processed 5M+ daily banking transactions at IDFC First Bank. Proficient in Python, SQL, PySpark, GCP, AWS, and Azure, with a focus on data quality, reconciliation, and pipeline reliability.

Technical Skills

Programming Languages: Python, SQL, PySpark, Scala, Java, PL/SQL, Bash/Unix Shell Scripting
Big Data & Processing: Apache Spark 2.x/3.x, Spark SQL, Spark DataFrames, Spark Streaming, Hadoop, HDFS, Hive, YARN, Databricks
Data Engineering: ETL/ELT Pipelines, Batch & Streaming Processing, Medallion Architecture, CDC, SCD Type 1 & 2, Data Modeling, Dimensional Modeling, Source-to-Target Mapping, Data Lineage
Orchestration & Streaming: Apache Airflow, Cloud Composer, Azure Data Factory, Apache Kafka, Google Pub/Sub, REST APIs, FastAPI
Cloud Platforms: GCP (BigQuery, Dataproc, Dataflow, Pub/Sub, Cloud Storage, Cloud Composer), AWS (S3, RDS, EC2, CloudWatch), Azure (ADLS, Data Factory, Synapse, Databricks)
Databases: BigQuery, Oracle, PostgreSQL, MySQL, SQL Server, MongoDB, Cassandra, DynamoDB
Data Quality & Monitoring: Data Validation, Data Reconciliation, Schema Validation, Data Profiling, Anomaly Detection, Cloud Logging, Cloud Monitoring, Azure Monitor, Splunk, Grafana
DevOps & Tools: Git, GitHub, Bitbucket, Jenkins, Maven, Docker, Kubernetes, Azure DevOps, CI/CD, Jira, Confluence
Visualization & Methodologies: Power BI, Tableau, Agile, Scrum

Professional Experience

Google Data Operations Center — May 2025 – Present

Data Engineer — *Omaha, NE*

- Build PySpark and Spark SQL pipelines on Dataproc processing ∼50 GB of daily data center operations data, including hardware asset inventory, maintenance work orders, and equipment telemetry.
- Develop Dataflow and BigQuery ELT pipelines integrating 12 source systems into curated datasets used by operations, capacity planning, and reporting teams.
- Orchestrate 40+ Cloud Composer (Airflow) DAGs with dependency management, retries, and SLA alerts, maintaining 99% on-time data delivery.
- Implemented Pub/Sub streaming ingestion for asset and ticket status events, reducing data availability lag from 4 hours to under 15 minutes.
- Optimized BigQuery tables with partitioning and clustering, reducing query data scanned and cost by 35%.
- Tuned Spark jobs through partition sizing, broadcast joins, and caching, cutting average job runtime from 45 to 18 minutes.
- Built a reusable Python, PySpark, and SQL validation framework covering 60+ tables to detect missing, duplicate, and late-arriving records before downstream use.
- Performed source-to-target reconciliation across 4 processing stages, identifying and resolving variances in 500K+ records per cycle.
- Set up Cloud Logging and Cloud Monitoring alerts for pipeline failures, reducing mean time to resolution by 30%.
- Detected schema and configuration drift across dev, test, and production environments, reducing data-promotion defects by 25%.
- Contribute to CI/CD deployments using Git, Jenkins, Maven, and Docker, and document lineage, mappings, and root-cause findings for stakeholders.

U.S. Renal Care — August 2023 – April 2025

Data Engineer — *NewYork, NY*

- Built Python, PySpark, and Airflow ETL/ELT pipelines consolidating dialysis treatment, patient census, scheduling, and clinic operations data from 300+ clinics into a centralized analytics warehouse.
- Ingested structured and semi-structured data from clinical systems, billing platforms, APIs, and AWS S3 into PostgreSQL, processing 15M+ records monthly.
- Designed data models for treatment sessions, patient encounters, and clinic staffing to support KPIs such as missed treatments, census trends, and clinic utilization.

- Reconciled treatment records against billing and claims data, identifying 200+ discrepancies per month and supporting revenue cycle accuracy.
- Implemented SQL and Python validation checks for PHI datasets with role-based access and masking, supporting HIPAA-compliant data handling.
- Developed parameter-driven transformation logic that reduced onboarding time for new clinic data feeds from 10 days to 3 days.
- Optimized SQL queries and indexing in PostgreSQL and MySQL, reducing report refresh time from 3 hours to 20 minutes.
- Monitored pipelines with AWS CloudWatch and resolved failed or delayed loads, maintaining a 98% pipeline success rate.
- Automated recurring validation and file-handling tasks with Python and Bash, saving 10+ hours of manual work per week.
- Documented data lineage, schemas, and transformation rules across ingestion, transformation, and reporting layers to improve traceability.

IDFC First Bank **July 2020 – July 2022**

Data Engineer *India*

- Built ETL workflows integrating customer, account, and loan data from legacy core banking and lending systems into a unified reporting layer.
- Processed 5M+ daily digital banking transactions (UPI, IMPS, NEFT) using Spark, Hive, and SQL on HDFS for reporting and analytics.
- Developed Oracle PL/SQL procedures for loan eligibility rules, EMI calculations, and validation of data used in regulatory reports.
- Designed customer and account data models and source-to-target mappings supporting a consolidated customer view across retail products.
- Reconciled transactions between core banking and downstream reporting datasets, resolving 95% of daily variances within SLA.
- Reduced nightly batch processing time from 6 hours to 3.5 hours by optimizing SQL, partitioning Hive tables, and restructuring job dependencies.
- Automated data validation and operational reports with Python and Bash, eliminating 12+ hours of manual effort per week.
- Supported AWS workflows using S3, EC2, and CloudWatch for data storage, processing, and monitoring.
- Investigated failed and delayed batch loads through log analysis and worked with QA and business teams to resolve production issues.

Projects

Pattern Mining for Multi-Disease Detection | *Python, R, Pandas, mlxtend, SciPy, Plotly* **May 2023 – Aug 2023**

- Engineered a preprocessing pipeline to unify heterogeneous clinical records, handling missing values, discretizing continuous lab measurements, and encoding patient profiles into transactional format for pattern mining.
- Performed statistical analysis (chi-square tests, correlation analysis) to identify significant co-occurrence signals among symptoms, risk factors, and diagnoses.
- Applied Apriori and FP-Growth algorithms to extract frequent itemsets and association rules, tuning support, confidence, and lift thresholds and benchmarking both algorithms for runtime and scalability.
- Pruned redundant rules and surfaced high-lift comorbidity patterns indicative of multiple co-occurring diseases to support early risk screening.
- Built interactive rule-network graphs and heatmaps in Plotly to make findings interpretable for healthcare decision-making.

Spoken Language Identification | *Python, TensorFlow, Keras, Librosa, ResNet-50, InceptionV3* **Jan 2023 – May 2023**

- Developed an audio processing pipeline with Librosa covering resampling, silence trimming, and normalization, and extracted MFCCs and log-Mel spectrograms as model inputs.
- Applied data augmentation (noise injection, time shifting, SpecAugment-style masking) to improve generalization across speakers and recording conditions.
- Trained a custom CNN baseline and fine-tuned ImageNet-pretrained ResNet-50 and InceptionV3 via transfer learning, using learning-rate scheduling and early stopping to optimize performance.
- Evaluated models using accuracy, macro F1-score, and confusion matrices to diagnose misclassification between acoustically similar languages.
- Built TensorBoard dashboards to monitor training and used Grad-CAM to visualize spectrogram regions driving model predictions.

Diabetic Retinopathy Severity Classification | *Python, R, OpenCV, scikit-learn, Seaborn* **Aug 2022 – Dec 2022**

- Conducted exploratory data analysis on retinal fundus images to study class distribution across severity grades and identify patterns in retinal abnormalities.
- Implemented image preprocessing with OpenCV (green-channel extraction, CLAHE contrast enhancement, normalization) to highlight microaneurysms and hemorrhages.
- Extracted statistical and texture features (intensity histograms, GLCM) and applied PCA and t-SNE for dimensionality reduction and cluster visualization.
- Trained and compared SVM, Random Forest, and XGBoost classifiers with stratified cross-validation, addressing class imbalance via class weighting and evaluating with quadratic weighted kappa.
- Built visual reports of model performance and feature importance to support clinical interpretation and guide further research.

Education

SUNY Polytechnic Institute **August 2022 – May 2024**

Master of Science in Computer and Information Science *Utica, NY*

Sreenidhi Institute of Science and Technology **2017 – 2021**

Bachelor of Technology in Electronics and Computer Engineering *Hyderabad, India*